

Notes (Week 4 Monday)

These are the notes I wrote during the lecture.

If $n + 1 = m + 1$ then $n = m$

If a set A

(i) $0 \in A$

(ii) If $n \in A$, then $S(n) \in A$

then $\forall n. n \in A$

If we replace the set with a predicate,
and successor arithmetic with "normal"
arithmetic, we get

If a set P

(i) $P(0)$

(ii) If $P(n)$, then $P(n+1)$

then $\forall n. P(n)$

Often, you can switch at will between
sets and predicates.

A set is uniquely defined by its
membership predicate.

A predicate can be viewed as the set
of all things that satisfy it.

If $\text{my_fun}(x)$ returns x , (IH)

then $\text{my_fun}(x+1)$ returns $x+1$

by definition of the function,

$\text{my_fun}(x+1)$ returns $x+1$

iff

$\text{my_fun}(x+2)$ returns $x+2$

$(\mathbb{N}, >)$

.. $7 > 6 > 5$

$(\mathbb{R}, <)$

$$\dots < 1/8 < 1/4 < 1/2 < 1$$
 $(\mathbb{Z}, <)$
 $(\mathbb{N}, <)$

$$\dots < 30 < 31$$

^ the length of this chain
is at most 31

 $(\mathcal{P}(\mathbb{N}), \subseteq)$
 $\{1, 2\} \subseteq \{1, 2\}$
 $z(\text{hello}) = 00000$

```
open bossLib;
```

```
Definition z_def:
```

```
  z [] = [] ∧
  z [a] = [0] ∧
  z (a::b::w) =
  0::z(0::w)
```

```
End
```

```
Theorem z_zeroises:
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```
  ∀w. z(w) =
    REPLICATE (LENGTH w) 0
```

```
Proof
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```
  Induct_on 'w'
  >- rw[z_def]
  >- (Cases_on 'w'
    >- (gvs[z_def] \EVAL_TAC)
    rw[z_def] >>
    Cases_on 't' >>
    gvs[z_def])
```

```
QED
```

$x + 1 > 5 \wedge p = 3$ $\leftarrow$ this week, we don't have that

$p \wedge q$ $\leftarrow$ no term language, instead, we just have
connectives ($\wedge, \vee$) and boolean variables
(more proper: propositional letters)

According to the first slide,

```
p ∧ ¬q  <- that's not a wff :(
(p ∧ ¬q) <- now it's a wff
```

$p \wedge q \wedge r$ might be written $p \cdot q \cdot r$ or pqr

$p \rightarrow q \rightarrow r$

$(p \rightarrow q) \rightarrow r$

is *not* equivalent to

$p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

$(T, \wedge, \vee, ', 0, 1)$ <-- when you see this in the Boolean Algebra slide,

$\wedge, \vee, ', 0, 1$ are not fixed. rather, think of them as variables names denoting functions over T .
And by choosing a specific T , and specific functions, we make a particular instance of
(the abstract concept of) a boolean algebra.

$\&\&$	vs	$\wedge$ (conjunction)
$\wedge$		$\wedge$
computes a truth value		given two <i>formulas</i> , constructs a new formula

$\mathcal{U} = \{\emptyset\}$

$T = \text{Pow}(\mathcal{U}) = \{\emptyset, \{\emptyset\}\}$

Think of $\emptyset$ as false,
and $\{\emptyset\}$ as true

In W2, we stated something like:

If you prove an equality on sets

$S = T$

with the laws of set operations,
you get
 $\text{dual}(S) = \text{dual}(T)$
for free

One-element boolean algebra

$(\{x\}, (s, t) \mapsto x, (s, t) \mapsto x, s \mapsto x, x, x)$

^ this boolean algebra satisfies

$$0 = 1$$

$$x \wedge x = x$$

$$x \vee x = x$$

$p \wedge q \rightarrow \neg q$ <-- this formula is neither true or false
 it's just a piece of syntax:
 it's a word over a particular alphabet,
 or a parse tree

We can *assign* truth values to the formula if we decide what the values of p and q are.

A *valuation* (aka states, models) is a function of type

$\text{Prop} \rightarrow \mathbb{B}$ function from propositional letters
 into the two-element boolean algebra

$p \mid q \mid p \wedge q \mid \neg q \mid p \wedge q \rightarrow \neg q$

$\top \mid \top \mid \top \mid \perp \mid$ $\perp$

$\top \mid \perp \mid \perp \mid \top \mid$ $\top$

.....

^ if we can be bothered to complete the table,
 we get the value of the formula under every possible valuation. A valuation just corresponds to a particular row in the truth table

Truth values are unsatisfactory for two reasons:

- We'd like to be more formal than "draw a picture"
- They don't scale that well.

For a formula with n distinct prop. letters, you'll have 2^n rows in your truth table.

[[]] these are called "semantic brackets"
 it's all Dana Scott's fault

If I have a thing (a piece of syntax), then
 [[thing]] (pronounced "the denotation of thing"),
 is some mathematical object that gives meaning

to thing.

$\llbracket \phi \rrbracket v$ "the truth value of the formula ϕ under the valuation v "

Logical law:

$$\phi \rightarrow \psi \equiv \neg \phi \vee \psi$$

Suppose we have a valuation v such that

$$p \mapsto 1$$

$$q \mapsto 0$$

$$\llbracket p \wedge q \rightarrow \neg q \rrbracket v =$$

$$!\llbracket p \wedge q \rrbracket v \mid \mid \llbracket \neg q \rrbracket v =$$

$$!(\llbracket p \rrbracket v \ \&\& \ \llbracket q \rrbracket v) \mid \mid \ !\llbracket q \rrbracket v =$$

$$!(1 \ \&\& \ 0) \mid \mid \ !0 =$$

$$!(0) \mid \mid \ 1 =$$

$$1 \mid \mid \ 1 =$$

$$1$$

$p \wedge q \rightarrow \neg q$ is satisfiable

$p \rightarrow p$ is a *tautology*

$\perp$ is unsatisfiable

$p \wedge \neg p$ is unsatisfiable

(Trivial) observation: every tautology is satisfiable.

Q: is biimplication the negation of xor?

A: yes

$$\llbracket p \leftrightarrow q \rrbracket v = \llbracket p \rightarrow q \rrbracket v \ \&\& \ \llbracket q \rightarrow p \rrbracket v$$